

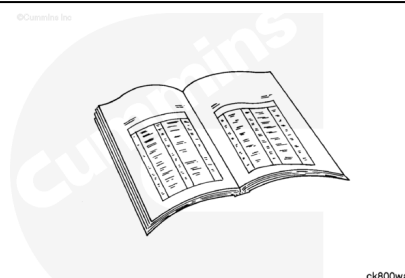
Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To prevent the possibility of personal injury, Always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

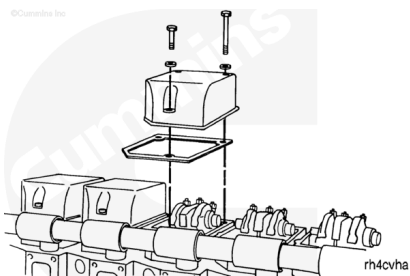
- Disconnect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Steam clean the engine to prevent dirt from entering the engine when the valve covers are removed. Refer to Procedure 000-009 (</qs3/pubsys2/xml/en/procedures/99/99-000-009.html>).
- Remove the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>).

Adjust

with Mechanically Actuated Injector

The Travel Method procedure can **only** be used to set fixed time PT® (type D) injectors.

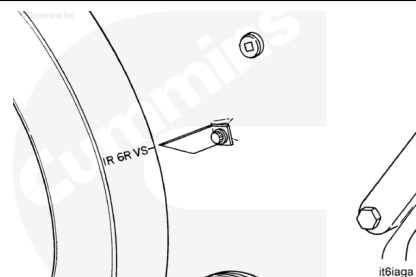
Remove the rocker lever covers and all related components.



The K38 and K50 engines have valve and injector adjustment marks in three locations.

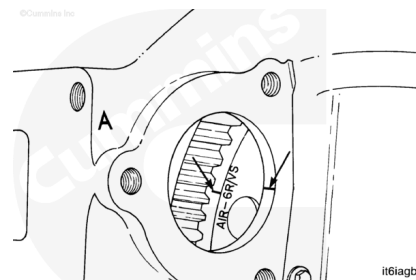
- Vibration damper
- Flywheel left bank (C)
- Flywheel right bank (A).

The marks on the vibration damper **must** be aligned with the pointer.



⚠ CAUTION ⚠

When using the flywheel index marks, the marks must begin with an "A", or the valves and injectors will not be adjusted correctly and can cause damage to the push rods.



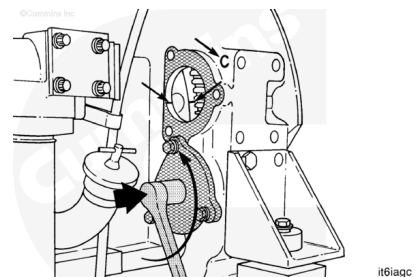
Right Bank

The starter bore cover **must** be removed to see the marks.

The engine barring device located on the right-bank of the engine **must** be used to rotate the flywheel.

⚠ CAUTION ⚠

When using the flywheel index marks, the marks must begin with an "C", or the valves and injectors will not be adjusted correctly and can cause damage to the push rods.



Left Bank

The starter bore cover **must** be removed to see the marks.

To use the barring device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.

The following is the firing order for all K38 engines.

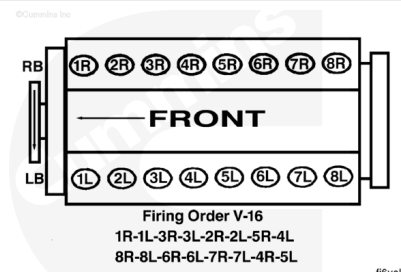
- 1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L

Note : Not all K50 engines have the same firing order. Engines have decals on the rocker lever covers and the engine dataplate is stamped with “REVISED FIRING ORDER”.

The following is the standard firing order for all K50 engines.

- 1R-1L-3R-3L-7R-7L-5R-5L-8R-8L-6R-6L-2R-2L-4R-4L

This graphic displays the “REVISED FIRING ORDER” for K50 engines.



The direction of normal rotation for the K38 and K50 engines is **clockwise** when viewing the front of the engine.

“VS” represents the valve to be set.

Ignore any “TC” (top center) marks while setting the valves and injectors.

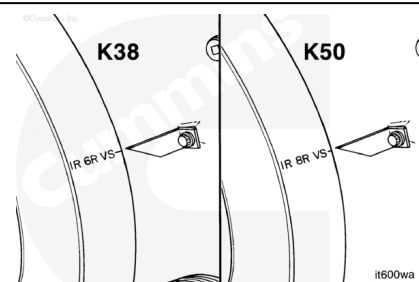
Use the barring device or a tool installed with crankshaft adapter capscrews to rotate the engine until the “VS” mark for the “R1” (number 1 right-bank cylinder) is aligned with the pointer or the index mark in the starter bore.

Determine the cylinder that is in position to set the valves:

The valves are ready to be adjusted on the cylinder that has all the valves closed.

Check the two cylinders shown on the “VS” mark.

The following steps assume that the rocker levers have just been installed and are **not** adjusted.

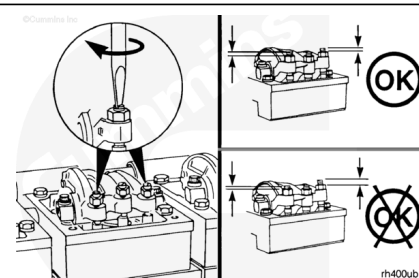


Perform this step on both cylinders to be checked.

All adjusting screws **must** be loose on all cylinders.

The push rod **must** remain in alignment.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods.



Turn the lock nuts until they touch the rocker levers. Compare the height of the adjusting screws above the lock nut.

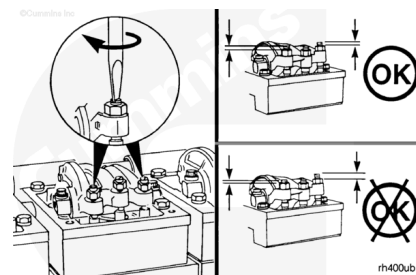
The cylinder that has the adjusting screws that are about the same height is ready for the valve adjustment.

Note : The push rods will be the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.

⚠ CAUTION ⚠

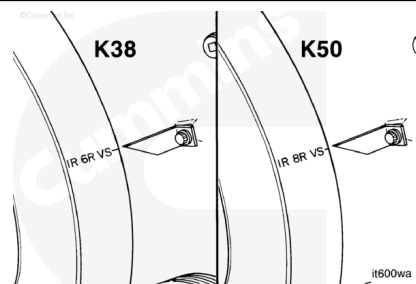
Use the correct valve and injector adjustment chart for the engine being serviced or parts can be damaged.

After identifying the cylinder with the valves ready to be adjusted, use the following charts for the sequence. The procedure and specifications for adjusting the valves and injectors are after the charts.



Note : Adjustment can begin on any valve set mark. In the illustration, assume the IR-6R or 1R-8R marks are aligned and the adjusting screw height for the valves on the cylinder number 1 right-bank are closed and ready to adjust.

The following charts give the crosshead, valve, and injector adjustment sequence.



K38 PT® (type D) Valve and Injector Adjustment Chart

	Valves Closed	Adjust Valves and Injectors
Valve Set Mark "VS" (L - Left-Bank) (R - Right-Bank)	On Cylinder Number (RB - Right-Bank) (LB - Left-Bank)	On Cylinder Number (RB - Right-Bank) (LB - Left-Bank)
1R-6R VS	1RB	1RB
6L-1L VS	6LB	6LB
5R-2R VS	5RB	5RB
2L-5L VS	2LB	2LB
3R-4R VS	3RB	3RB
4L-3L VS	4LB	4LB
1R-6R VS	6RB	6RB
6L-1L VS	1LB	1LB
5R-2R VS	2RB	2RB
2L-5L VS	5LB	5LB

K38 PT® (type D) Valve and Injector Adjustment Chart

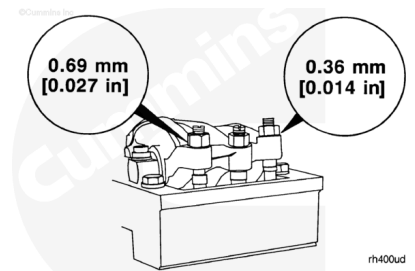
	Valves Closed	Adjust Valves and Injectors
3R-4R VS	4RB	4RB
4L-3L VS	3LB	3LB

K50 (type D) Valve and Injector Adjustment Chart

	Valves Closed	Adjust Valves and Injectors
Valve Set Mark "VS" (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (RB - Right-Bank) (LB - Left-Bank)	On Cylinder Number (RB - Right-Bank) (LB - Left-Bank)
1R-8R VS	1RB	1RB
1L-8L VS	1LB	1LB
3R-6R VS	3RB	3RB
3L-6R VS	3LB	3LB
2R-7R VS	7RB	7RB
2L-7L VS	7LB	7LB
4R-5R VS	5RB	5RB
4L-5L VS	5LB	5LB
1R-8R VS	8RB	8RB
1L-8L VS	8LB	8LB
3R-6R VS	6RB	6RB
3L-6L VS	6LB	6LB
2R-7R VS	2RB	2RB
2L-7L VS	2LB	2LB
4R-5R VS	4RB	4RB
4L-5L VS	4LB	4LB

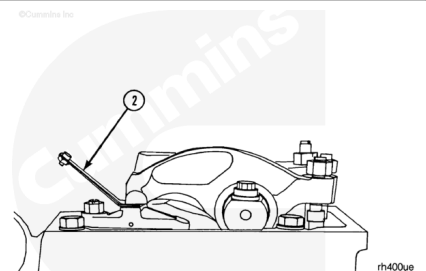
Valve Adjustment (Initial Set)

mm		in
0.69	Exhaust	0.027
0.36	Intake	0.014



Note : Some engines have guideless crossheads and do not require adjustment.

Select a feeler gauge for the correct valve lash specification. Insert the gauge (2) between the rocker lever and the crosshead.



Two different methods for establishing valve lash clearance are described below. Either method can be used. The Torque Wrench method has proven to be the most consistent.

Torque Wrench Method:

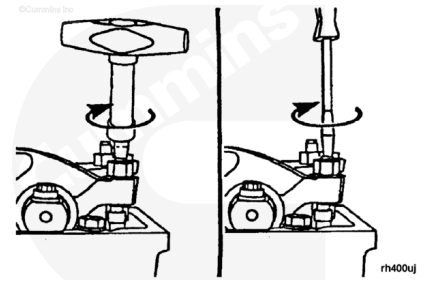
Use the inch-pound torque wrench, Part Number 3376592.

Tighten the adjusting screw.

Torque Value: 0.68 n•m [6 in-lb]

Feeler Gauge Method:

Use a screwdriver to turn the adjusting screw **only** until the lever touches the feeler gauge.



The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut.

With torque wrench adapter, Part Number 3163196 (1).

Torque Value: 45 n•m [35 ft-lb]

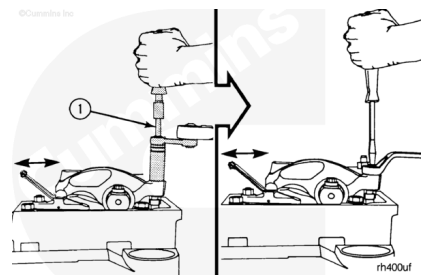
Without torque wrench adapter.

Torque Value: 60 n•m [45 ft-lb]

The feeler gauge **must** slide backward and forward with **only** a slight drag.

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the feeler gauge will fit.

Repeat the adjustment process until the clearance is correct on both the intake and exhaust valves for the cylinder being adjusted.

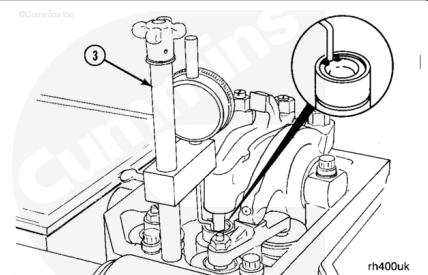


Injector Adjustment Indicator Method:

Assemble the parts of the injector and valve adjustment kit (3), Part Number 3822575, or equivalent.

Install the adjustment kit on the cylinder to be adjusted as shown.

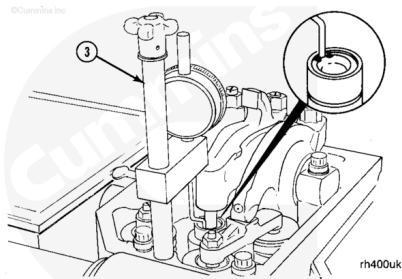
Adjust the indicator so that the tip is touching the top of the injector plunger.



The tip **must not** touch the rocker lever.

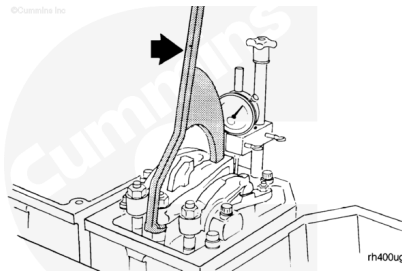
Lower the indicator 12.7 mm [0.50 in] to allow for travel.

Lock the indicator support to the post.



⚠ WARNING ⚠

The injector plunger is under spring tension. Allowing the tool to slip can cause personal injury.



⚠ CAUTION ⚠

To prevent damage to the indicator, allow the lever to return slowly.

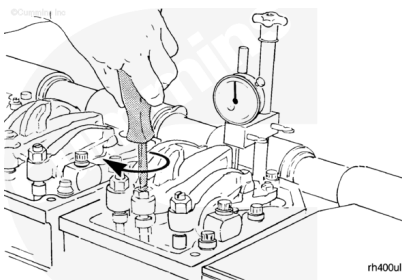
Use a rocker lever actuator, Part Number 3375539, or equivalent.

Depress the lever until the injector bottoms 2 or 3 times. This will remove all of the fuel and oil from the injector cup.

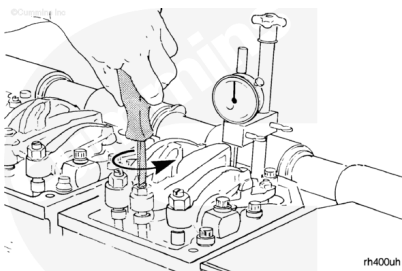
Tighten the adjusting screw until the injector plunger is bottomed in the cup.

Set the dial indicator to zero.

Turn the adjusting screw backward and forward 2 to 3 times to confirm the zero setting.



Loosen the adjusting screw until the indicator reads the specified travel.



Measurements

	mm	in
Injector Travel	7.811	0.3075

The adjusting screw **must not** turn when the lock nut is tightened.

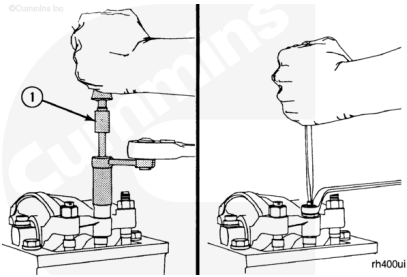
Tighten the lock nut.

With torque wrench adapter, Part Number 3163196 (1).

Torque Value: 45 n•m [35 ft-lb]

Without torque wrench adapter.

Torque Value: 60 n•m [45 ft-lb]

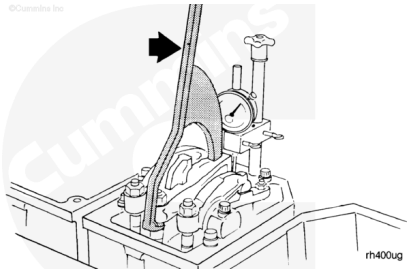


⚠ WARNING ⚠

The injector plunger is under spring tension. Do not allow the tool to slip or personal injury can result.

⚠ CAUTION ⚠

To prevent damage to the indicator, allow the lever to return slowly.



Check the injector adjustment.

Use the rocker lever actuator to bottom the injector plunger.
Confirm the zero setting on the indicator.

Allow the rocker lever to return slowly. Check the injector setting. Repeat the adjustment process if it is **not** within specification.

mm		in
7.80	MIN	0.307
7.82	MAX	0.308

Rotate the engine to align the next timing mark.

Adjust the appropriate valves and injectors.

Repeat the process to adjust all of the valves and the injectors correctly.

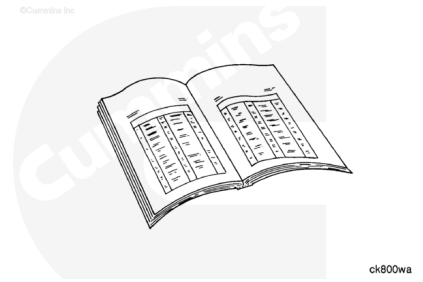
If the barring device was used, allow the spring to push the shaft and clear the ring gear.

Install the clip.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To prevent the possibility of personal injury, Always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the rocker lever cover. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/28/28-003-001-tr.html>).
- Connect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Operate the engine and check for leaks.